Python Program Execution & Test Cases

Script: one.py **Module:** Class Marks Analysis Report **Environment:** PowerShell (Python 3)

Test Case 1: Multi-Student Dataset

Input: 40 50 60 70 80 — Verifies calculation of statistical indicators (Mean, Median, Std Dev, Pass rate, Above-Average count) with 100% passing results.

```
PS D:\Aakesh\Project> python3 one.py
Enter marks separated by spaces: 40 50 60 70 80

--- Class Marks Analysis Report ---
Mean                            : 60.00
Median                          : 60.00
Maximum                         : 80.00
Minimum                         : 40.00
Population Standard Deviation   : 14.14
Passed Students                 : 5
Failed Students                 : 0
Pass Percentage                 : 100.00%
Above-Average Students          : 2
PS D:\Aakesh\Project>
```

Test Case 2: Single Entry

Input: 75 — Verifies single element handling (Mean=75, Std Dev=0.00).

```
PS D:\Aakesh\Project> python3 one.py
Enter marks separated by spaces: 75

--- Class Marks Analysis Report ---
Mean                            : 75.00
Median                          : 75.00
Maximum                         : 75.00
Minimum                         : 75.00
Population Standard Deviation   : 0.00
Passed Students                 : 1
Failed Students                 : 0
Pass Percentage                 : 100.00%
Above-Average Students          : 0
PS D:\Aakesh\Project>
```

Test Case 3: Mixed Pass/Fail Distribution

Input: 25 50 75 100 — Evaluates passing threshold with 1 failed and 3 passed students.

```
PS D:\Aakesh\Project> python3 one.py
Enter marks separated by spaces: 25 50 75 100

--- Class Marks Analysis Report ---
Mean                            : 62.50
Median                          : 62.50
Maximum                         : 100.00
Minimum                         : 25.00
Population Standard Deviation   : 27.95
Passed Students                 : 3
Failed Students                 : 1
Pass Percentage                 : 75.00%
Above-Average Students          : 2
PS D:\Aakesh\Project>
```

Test Case 4: Non-Numeric Validation

Input: 50 abc 70 — Validates character error handling (*Invalid marks 'abc'. Enter numbers only.*).

```
PS D:\Aakesh\Project> python3 one.py
Enter marks separated by spaces: 50 abc 70
Invalid marks 'abc'. Enter numbers only.
PS D:\Aakesh\Project>
```

Test Case 5: Out of Range Check

Input: 50 105 70 — Validates numeric boundaries (*Marks must be between 0 and 100.*).

```
PS D:\Aakesh\Project>  python3 one.py
Enter marks separated by spaces: 50 105 70
Marks must be between 0 and 100.
PS D:\Aakesh\Project>
```